



Gemini 3 Pro Image

Model Card

Gemini 3 Pro Image - Model Card

Model Cards are intended to provide essential information on Gemini models, including known limitations, mitigation approaches, and safety performance. Model cards may be updated from time-to-time; for example, to include updated evaluations as the model is improved or revised. See the [Google DeepMind site](#) for a comprehensive list of model cards.

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Model Information

Description: Gemini 3 Pro Image is the next generation in the Gemini series of models, a suite of highly-capable, natively multimodal, reasoning models. Gemini 3 Pro Image is now Google's most advanced model for image generation and can comprehend vast datasets, challenging problems from different information sources, including text and images.

Model dependencies: Gemini 3 Pro Image is based on Gemini 3 Pro.

Inputs: Text strings (e.g. a question, a prompt, document(s) to be summarized), images, with a token context window of up to 1M.

Outputs: Image, with a 64K token output

Architecture Gemini 3 Pro Image is based on Gemini 3 Pro. For more information about the model architecture for Gemini 3 Pro, see the Gemini 3 Pro [model card](#).

Model Data

Training Dataset: Gemini 3 Pro Image is based on Gemini 3 Pro. For more information about the training dataset for Gemini 3 Pro Image, see the Gemini 3 Pro [model card](#).

Training Data Processing: For more information about the training data processing for Gemini 3 Pro Image, see the Gemini 3 Pro [model card](#).

Implementation and Sustainability

Hardware: Gemini 3 Pro Image was trained using [Google's Tensor Processing Units](#) (TPUs). TPUs are specifically designed to handle the massive computations involved in training LLMs and can speed up training considerably compared to CPUs. TPUs often come with large amounts of high-bandwidth memory, allowing for the handling of large models and batch sizes during training, which can lead to better model quality. TPU Pods (large clusters of TPUs) also provide a scalable solution for handling the growing complexity of large foundation models. Training can be distributed across multiple TPU devices for faster and more efficient processing.

The efficiencies gained through the use of TPUs are aligned with Google's [commitment to operate sustainably](#).

Software: Training was done using [JAX](#) and [ML Pathways](#).

Distribution

Gemini 3 Pro Image is based on Gemini 3 Pro. For more information about the distribution for Gemini 3 Pro Image, see the Gemini 3 Pro [model card](#).

Evaluation

The following Evaluation approach and results are for Gemini 3 Pro Image. For more information about the evaluation for Gemini 3 Pro, see the Gemini 3 Pro [model card](#).

Approach: Gemini 3 Pro Image was evaluated using the methodology below:

- **Human evaluations** of several different quality aspects of image generation were conducted. Prompt sets for human evaluations were in two broad categories:
 - **New capability sets:** diverse T2I and editing evals are curated, covering a wide range of capabilities, e.g. text, style, character, reasoning, factuality, infographics, ink (doodle) based editing, multi-turn, multi-image (visual design, virtual try on, product recontextualization, etc.)
 - **Regression sets:** popular use cases observed on Gemini 2.5 Flash Image, to ensure Gemini 3 Pro Image does not show noticeable regression.
- **Capabilities / Benchmarks:** human evaluations were conducted on these benchmarks:
 - **Existing Capabilities:** Text Rendering, Stylization, Multi-Turn, General Image Editing, Character Editing, Object/Environment Editing, Generate Text-to-Image and Regressions.
 - **New Capabilities:** Multi character editing, Chart editing, Text editing, Factuality - Edu, Multi-input, Infographics, Doodle editing and Visual design.

Results: Gemini 3 Pro Image demonstrated strong performance and leads on Elo score across all new capabilities and regression sets. It leads with significant margins in complex tasks, e.g. multi-character, chart and text editing. It improves substantially in style compared to Gemini 2.5 Flash Image.

Existing Capabilities:

Capability Benchmark	Gemini 3 Pro Image (Search On)	Gemini 2.5 Flash Image	GPT-Image 1	Seedream v4 4k	Flux Pro Kontext Max
Text Rendering	1198 ± 18	997 ± 10	1150 ± 14	1019 ± 13	854 ± 13
Stylization	1098 ± 11	933 ± 7	1069 ± 9	991 ± 9	908 ± 11
Multi-Turn	1186 ± 19	1045 ± 24	1079 ± 32	990 ± 32	889 ± 37
General Image Editing	1127 ± 13	996 ± 8	1011 ± 13	965 ± 12	902 ± 13
Character Editing	1176 ± 16	1075 ± 8	1016 ± 10	889 ± 10	843 ± 10
Object/Env. Editing	1102 ± 19	1025 ± 9	930 ± 12	983 ± 13	961 ± 10
General Text-to-Image	1094 ± 16	1037 ± 8	1025 ± 9	1011 ± 9	907 ± 9
Regression sets / popular use cases	1128 ± 19	1086 ± 9	1039 ± 13	900 ± 11	847 ± 14

New Capabilities:

Capability Benchmark	Gemini 3 Pro Image (Search On)	Gemini 2.5 Flash Image	GPT-Image 1	Seedream v4 4k	Flux Pro Kontext Max
Multi character editing	1213 ± 16	950 ± 10	997 ± 13	840 ± 19	-
Chart editing	1209 ± 18	971 ± 10	994 ± 16	934 ± 16	893 ± 15
Text editing	1202 ± 23	1001 ± 10	996 ± 14	860 ± 15	943 ± 12
Factuality - Edu	1169 ± 25	1050 ± 11	1084 ± 25	969 ± 22	884 ± 26
Multi-input 1-3	1109 ± 16	993 ± 8	970 ± 12	928 ± 11	-
Infographics	1268 ± 17	1162 ± 11	1087 ± 12	1049 ± 12	824 ± 15
Doodle editing	1106 ± 14	1038 ± 7	955 ± 9	1017 ± 9	884 ± 9
Visual design	1104 ± 16	1083 ± 7	1028 ± 11	1038 ± 12	907 ± 11

Intended Usage and Limitations

Benefit and Intended Usage: Gemini 3 Pro Image is our most powerful AI image editing model yet, capable of using Gemini's real-world knowledge to deliver precise results, from complex infographics to historically accurate scenes. It is well-suited for applications that require:

- creation and editing of images with professional levels of precision and control
- generation of clear text for posters and intricate diagrams
- long context real-world knowledge
- studio-quality control

Known Limitations: Gemini 3 Pro Image may exhibit some of the general limitations of foundation models, such as hallucinations. There may also be occasional slowness or timeout issues.

Even though it leads across capabilities compared to other models, Gemini 3 Pro Image still has room for several quality improvements:

- Text rendering: poor in small text (often blurry in 1k model), long paragraphs, page length
- Character consistency is not always perfect between input images and generated output image
- Masked/Doodle based editing: partial instruction following and persistent ink
- When editing images: infrequent copying/pasting from user's input image to generated image

- Occasional confusion around spatial localisation (e.g. left/right etc.)
- Still limited in advanced capabilities with world knowledge, 3D reasoning and factuality

The knowledge cutoff date for Gemini 3 Pro Image was January 2025.

Acceptable Usage: For more information about the acceptable usage for Gemini 3 Pro Image, see the Gemini 3 Pro [model card](#).

Ethics and Content Safety

Evaluation Approach: Gemini 3 Pro Image was developed in partnership with internal safety, and responsibility teams. A range of evaluations and red teaming activities were conducted to help improve the model and inform decision-making. These evaluations and activities align with [Google's AI Principles](#) and [responsible AI approach](#), as well as Google's Generative AI policies (e.g. [Gen AI Prohibited Use Policy](#) and the [Gemini API Additional Terms of Service](#)). As Gemini 3 Pro Image is based on Gemini 3 Pro, see the Gemini 3 Pro [model card](#) for additional Ethics & Content Safety details.

Evaluation types included but were not limited to:

- **Training/Development Evaluations** including automated and human evaluations carried out continuously throughout and after the model's training, to monitor its progress and performance;
- **Human Red Teaming** conducted by specialist teams across the policies and desiderata, deliberately trying to spot weaknesses and ensure the model adheres to safety policies and desired outcomes;
- **Trust Assurance Evaluations** conducted by evaluators who sit outside of the model development team, used to independently assess responsibility and safety governance decisions;
- **Ethics & Safety Reviews** were conducted ahead of the model's release.

In addition, we perform testing following the guidelines in [Google DeepMind's Frontier Safety Framework](#) (FSF).

Safety Policies: Gemini's safety policies are based on Google's standard framework, which aim to prevent our Generative AI models from generating harmful content, including:

1. Content related to child sexual abuse material and exploitation
2. Hate speech (e.g. dehumanizing members of protected groups)
3. Dangerous content (e.g. promoting suicide, or instructing in activities that could cause real-world harm)
4. Harassment (e.g. encouraging violence against people)
5. Sexually explicit content

6. Medical advice that runs contrary to scientific or medical consensus

We continue to improve our internal evaluations, including refining automated evaluations to reduce false positives and negatives, as well as update query sets to ensure balance and maintain a high standard of results.

Frontier Safety Assessment: We evaluated Gemini 3 Pro for Frontier Safety and reported the results in the [Gemini 3 Pro Frontier Safety Framework Report](#), finding that it did not reach any critical capability levels outlined in our Frontier Safety Framework.

Risks and Mitigations: Safety and responsibility was built into Gemini 3 Pro Image throughout the training and deployment lifecycle, including pre-training, post-training, and product-level mitigations. Mitigations include, but are not limited to:

- dataset filtering;
- conditional pre-training;
- supervised fine-tuning;
- reinforcement learning from human and critic feedback;
- safety policies and desiderata;
- product-level mitigations such as safety filtering.